

Investigation into the Effects of Autonomy and Emotional Environments on Sense of Agency in VR

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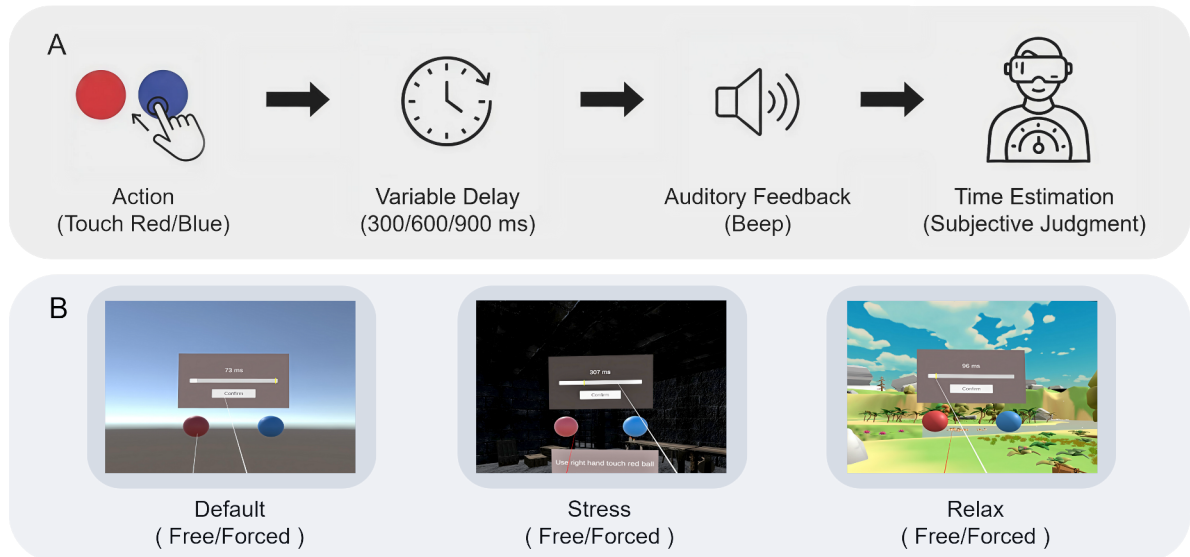


Figure 1: A. Experiment Flowchart, B. Experiment Scenes

ABSTRACT

Sense of Agency (SoA) is vital for VR user experience, yet the interplay between environmental emotion and autonomy remains unclear. This study investigates how free choice vs. forced action impacts SoA across three emotional contexts (stressful, relaxing, neutral) using Intentional Binding, the SoA Scale, and PANAS. Results show free choice significantly enhances both implicit and explicit SoA. Notably, a marginally significant interaction suggests stressful environments may amplify SoA loss when autonomy is deprived. These findings reveal how environmental emotions regulate perceived control, offering theoretical guidance for balancing atmosphere and user agency in immersive VR design.

Index Terms: Sense of Agency, Virtual Reality, Human-Computer Interaction, Intentional Binding.

1 INTRODUCTION

The Sense of Agency (SoA) refers to an individual's subjective perception of being the initiator of actions and having control over

external events [1]. In VR systems, SoA is a core factor in assessing the Quality of Interaction and user immersion. Enhancing SoA helps reduce sensorimotor conflicts and strengthens users' sense of control over the virtual environment.

Although previous studies have indicated that users' ability to independently determine interactive behaviors (i.e., free choice) is generally regarded as a critical prerequisite for the emergence of SoA[2], most existing VR research has been conducted in neutral laboratory environments. Practical VR application scenes often involve complex visual feedback and emotional induction. Currently, high-resolution display technologies and low-latency tracking systems provide the material basis for creating highly immersive emotional atmospheres. However, there is a lack of systematic quantitative evidence regarding how such strong emotional induction characteristics regulate users' cognitive processes under different autonomy levels.

This study employs the HTC Vive system to quantitatively analyze the interactive effects of autonomy on SoA across three typical VR scenes through Intentional Binding [3] (implicit measurement) and subjective scales (explicit measurement).

2 EXPERIMENT

To investigate the interplay between emotional context and autonomy, we conducted a 3 (Environment: Default/Relax/Stress) $\times$ 2 (Choice: Free vs. Forced) experiment.

2.1 Participants & Platform

Nineteen healthy adults (9 males and 10 females, $M = 22.47$, $SD = 1.26$) participated. We utilized the HTC VIVE Pro for immersive display and Vive Controller 2.0 for precise hand-tracking and interaction consistency.

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2.2 Design & Procedure

Three Unity3D scenes were designed: Default (neutral gray), Relax (rural landscape), and Stress (dim, enclosed space). Two virtual balls (a red and a blue) were placed in the center of the scene (1m apart, 1.2m high). In each trial, participants were allowed to touch either ball with either hand under Free choice condition or required to comply random visual instructions under Forced action conditions (e.g., “Touch the red ball with your left hand”). Auditory feedback followed after a random delay (300/600/900 ms).

2.3 Measurement Tools

SoA was assessed via Intentional Binding (implicit: estimating interval between touch and sound. Shorter time estimates indicate stronger SoA) and a 13-item SoA Scale (explicit 7-point Likert) derived from the research of Tapal [4]. Implicit SoA was measured via interval estimation instead of the Libet clock to avoid visuospatial attentional biases [5]. PANAS scores validated the emotional induction of each scene [6].

3 RESULTS

3.1 Statistical Analysis

Linear Mixed Models (LMM) were used to analyze the data. “Environment” and “Autonomy” were set as fixed effects, and “Subject ID” was included as a random effect to control for individual differences. All LMMs included “Delay” as a covariate. Bonferroni correction was applied for post-hoc multiple comparisons.

3.2 Validation of Emotional Induction Effectiveness (PANAS)

The results confirmed that the VR scenes successfully induced the expected emotions. For the Positive Affect (PA) scale, a significant main effect of environment was observed ($F_{(2,90)} = 11.04$, $p < .001$). Post-hoc comparisons (Bonferroni) revealed that PA scores in the Relax scene were significantly higher than those in the Default scene, while PA scores in the Stress scene were significantly lower than those in the Default scene. For the Negative Affect (NA) scale, a significant main effect of environment was also found ($F_{(2,90)} = 9.09$, $p < .001$). NA scores in the Stress scene were significantly higher than those in the other two scenes, and the Relax scene induced the lowest negative emotions. These results quantitatively confirm the effectiveness of the VR emotional induction task.

3.3 Intentional Binding Effect - Core Behavioral Indicator

LMM analysis showed an extremely significant main effect of autonomy ($F_{(1,5105)} = 39.675$, $p < .001$). The estimated time in the Free condition ($M = 425.89$ ms) was significantly shorter than that in the Forced condition ($M = 457.27$ ms). This indicates that free choice enhances the perceived time compression between actions and feedback, i.e., stronger implicit SoA. A marginally significant interaction trend was observed between environment and autonomy ($F_{(2,5105)} = 2.964$, $p = .052$). Notably, the stressful environment exhibited a potential “magnifying effect”: under the Stress environment, the perceived difference caused by the loss of autonomy (i.e., reduced time compression, $\Delta = 43.76$ ms) was nearly three times that in the neutral environment ($\Delta = 14.90$ ms).

3.4 Subjective SoA Ratings (SoA Scale)

The results of the subjective scale were highly consistent with the behavioral data. A significant main effect of autonomy was found ($F_{(1,90)} = 18.39$, $p < .001$). SoA ratings in the Free condition ($M = 5.51$, $SE = 0.33$) were significantly higher than those in the Forced condition ($M = 4.84$, $SE = 0.33$).

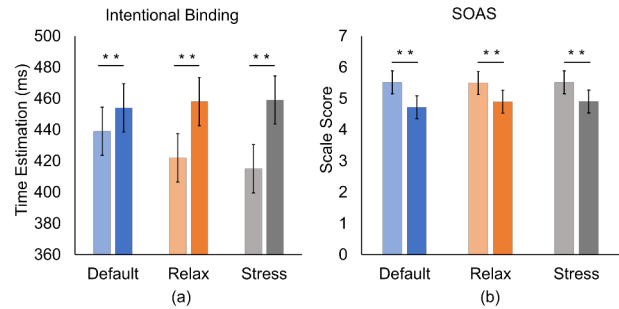


Figure 2: (a)Implicit SoA (Intentional Binding), (b) Explicit SoA (SoA Scale). Lighter colors indicate “Free” condition and darker colors indicate “Forced” condition ($\alpha = .05$).

4 DISCUSSION

Autonomy as the Core Driver: Both implicit and explicit measures confirm that free choice significantly enhances SoA. Allowing users to independently initiate actions reduces cognitive friction and improves interaction coherence in VR.

Validated Emotional Induction: PANAS results confirm that visual-based VR atmospheres effectively regulate psychological states. It ensures that the observed fluctuations in SoA are indeed caused by environment-induced changes in psychological states rather than accidental factors.

The “Magnifying Effect” of Environmental Stress on SoA Loss: This study also found a marginally significant interaction between environmental emotions and autonomy ($p = .052$). This may indicate that environmental stress heightens sensitivity to autonomy loss. While users may tolerate system-mandated actions in neutral/relaxing settings, stressful environments may amplify the cognitive dissonance of being “forced”.

5 CONCLUSION

This study explores how emotional contexts in VR environments influence users’ SoA under different autonomy levels. The experimental results indicate that free choice is a core driver of SoA, and stressful environments may amplify the negative cognitive effects of reduced autonomy. Future work will incorporate auditory induction to further reveal the magnitude and mechanism of this interactive process.

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